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# Conditioned hydrogen for a green hydrogen supply for heavy duty-vehicles in 2030 and 2050 – A techno-economic well-to-tank assessment of various supply chains

*Lucas Sens* [a,*], *Ulf Neuling* [a], *Karsten Wilbrand* [b], *Martin Kaltschmitt* [a]

[a] *Institute of Environmental Technology and Energy Economics, Hamburg University of Technology, Eissendorfer Straße 40, 21073, Hamburg, Germany*

[b] *Shell Global Solutions (Deutschland) GmbH, Hohe-Schaar-Straße 36, 21107, Hamburg, Germany*

## HIGHLIGHTS

- Well to tank analysis for a hydrogen supply of filling stations.
- Gaseous hydrogen supply chains are the lowest cost options for gaseous filling.
- Liquid hydrogen supply chains show the lowest supply costs for liquid filling.
- LOHC obtain high supply costs, especially for long transportation distances.

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## ABSTRACT

Hydrogen, seen as a promising secondary renewable energy carrier, can becompressed or liquefied as well as chemically bound to enable a more economic storage and transportation. All conditioned hydrogen options obtain specific advantages and disadvantages. Thus, this paper assess the energy efficiency and the cost of a green hydrogen supply from well to tank exemplified for fuel cell heavy-duty vehicles in Germany related to the years 2030 and 2050. Therefore, compressed gaseous hydrogen, liquid hydrogen, liquid organic hydrogen carriers, methanol and ammonia are considered and different hydrogen production regions in North and Central Germany, as well as an import from Tunisia and Argentina (Patagonia) are assumed. The assessment shows that a hydrogen supply chain in its state of aggregation as finally demanded is the lowest cost option leading to 700 bar compressed gaseous hydrogen filling costs of 5 $€_{2020}/kg_{H2}$ in 2030 and 4 $€_{2020}/kg_{H2}$ in 2050 and liquid hydrogen filling costs around 7 $€_{2020}/kg_{H2}$ in 2030 and 6 $€_{2020}/kg_{H2}$ in 2050.

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* *Corresponding author.*
E-mail address: lucas.sens@tuhh.de (L. Sens).

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**Abbreviation**

*Meaning*

$a_i$ Compressibility factor coefficient
$ACAPEX_i$ Annuity of capital expenditures of i, €$_{2020}$/a
ARG Hydrogen production in Argentina
$b_i$ Compressibility factor coefficient
$B_i$ Compressibility factor coefficient,
$c_{conversion}$ Conversion cost, €$_{2020}$/kg
$c_i$ Compressibility factor coefficient
$c_{fill}$ Hydrogen filling cost, €$_{2020}$/kg
$c_{LOHC}$ LOHC cost for hydrogen supply, €$_{2020}$/kg
$c_{overall}$ Overall hydrogen supply cost, €$_{2020}$/kg
$c_{production}$ Hydrogen production cost, €$_{2020}$/kg
$c_{reconverison}$ Reconversion cost, €$_{2020}$/kg
$c_{section,i}$ Chain section i cost, €$_{2020}$/kg
$c_{storage}$ Storage cost, €$_{2020}$/kg
$c_{transport}$ Transport cost, €$_{2020}$/kg
$C_i$ Compressibility factor coefficient
CAPEX Capital expenditure, €$_{2020}$
$CAPEX_i$ Capital expenditure of i, €$_{2020}$/a
$CAPEX_{i,base}$ Capital expenditure of base plant i, €$_{2020}$
$CAPEX_{i,design}$ Capital expenditure of design plant i, €$_{2020}$
$CAPEX_{i,inst}$ Capital expenditure of plant i including installation expenditures, €$_{2020}$
$CAPEX_{LOHC}$ Capital expenditure for LOHC material, €$_{2020}$
CGH2 Compressed gaseous hydrogen
CH3OH Methanol
CO2 Carbon dioxide
CT Cryotank
$Cycle_{LOHC,annual}$ LOHC round trip cycles per year, n/a
$d_i$ Depreciation time, a
$d_{LOHC}$ Depreciation time of LOHC material, a
$f_{inst}$ Installation factor
$f_{n,t}$ Capacity factor of n at time step t, %
FC Fuel cell
FCEV Fuel cell electric vehicle
GER Hydrogen production in North Germany
GHG Greenhouse gas
GH2 Gaseous hydrogen
GNH3 Gaseous ammonia
HDV Heavy duty-vehicle
INFL Inflation
IT Isolated tank
LH2 Liquid hydrogen
LHV Lower heating value
LNH3 Liquid ammonia
LOC Local hydrogen production in Central Germany
LOHC Liquid organic hydrogen carrier
$m_{H2,fill,annual}$ Annual hydrogen filled, kg/a
$m_{H2,prod,annual}$ Annual hydrogen produced, kg/a
$m_{H2,loss,annual}$ Annual hydrogen loss, kg/a
MOT Mineral oil tank
N2 Nitrogen
NH3 Ammonia
$OPEX_i$ Operational expenditures of i, €$_{2020}$/(kg a)
$p_{in}$ Inlet pressure, bar
$p_{out}$ Outlet pressure, bar
PEM Polymer electrolyte membrane
PT Pressure tank
PtX Power-to-X
$q_{comp}$ Compression energy input, kWh/kg
$q_{conversion}$ Conversion energy input, kWh/kg
$q_{fill}$ Filling energy input, kWh/kg
$q_{H2,nozzle}$ Hydrogen energy content at nozzle, kWh/kg
$q_{overall}$ Overall energy input, kWh/kg
$q_{production}$ Hydrogen production energy input, kWh/kg
$q_{production,ideal}$ Ideal hydrogen production energy input, kWh/kg
$q_{production,losses}$ Additional hydrogen production energy input, kWh/kg
$q_{reconversion}$ Reconversion energy input, kWh/kg
$q_{storage}$ Storage energy input, kWh/kg
$q_{transport}$ Transport energy input, kWh/kg
$Q_{el,bat,in}$ Battery electricity input, kWh
$Q_{el,bat,out}$ Battery electricity output, kWh
$Q_{el,comp,in}$ Compressor electricity input, kWh
$Q_{el,demand}$ Electricity demand, kWh
$Q_{el,pemel,in}$ Electrolyzer electricity input, kWh
$Q_{el,production}$ Electricity production, kWh
$Q_{el,supply}$ Electricity supply, kWh
$Q_{H2,n,in,t}$ Hydrogen input in storage n at time step t, kg
$Q_{H2,n,out,t}$ Hydrogen output from storage n at time step t, kg
$Q_{H2,pemel,out}$ Electrolyzer hydrogen output, kg
$Q_{H2,supply}$ Hydrogen supply, kg
$Q_{W,pemel}$ Electrolyzer power, kWh
$R_s$ Specific gas constant of hydrogen, J/(kg K)
$S_{i,base}$ Size of base plant, e.g. tpd
$S_{i,design}$ Size of design plant, e.g. tpd
$SOC_{H2,n,t}$ State of charge of hydrogen storage option n at time step t, %
$SOC_{H2,n,t+1}$ State of charge of hydrogen storage option n at time step t+1, %
$t_{LOHC,conversion}$ Time for LOHC conversion, h
$t_{LOHC,cycle}$ Time for one LOHC round trip cycle, h
$t_{LOHC,storage}$ Time for LOHC storage, h
$t_{LOHC,reconverison}$ Time for LOHC reconversion, h
$t_{LOHC,transport}$ Time for LOHC transport, h
$T_{comp}$ Compression temperature, K
TC Total cost, €$_{2020}$
$TLCC_n$ Total life cycle cost of n, €$_{2020}$/kW or €$_{2020}$/kWh
TUN Hydrogen production in Tunisa
$W_n$ Installed capacity or power of n, kW or kWh
$WACC_{nom}$ Nominal weighted averagecost of capital
$WACC_{real}$ Real weighted average cost of capital
$z_{in}$ Compressibility factor of hydrogen at intlet pressure
$z_{out}$ Compressibility factor of hydrogen at outlet pressure
$\alpha_i$ Scaling factor of i
$\eta_{bat,out}$ Battery output storage efficiency, %
$\eta_{Comp}$ Compressor efficiency, %
$\eta_{SC}$ Overall supply chain efficiency, %

## Introduction and background

The overarching goal of the Paris Agreement from 2015, where most nations of the world declared to limit the increase of the global mean temperature below 2 °C above pre-industrial levels, requires an energy system transition towards carbon-neutral energy sources. Due to the ongoing cost decline of electricity from wind and solar, electricity from such renewable sources and the subsequent Power-to-X (PtX) products like hydrogen are excepted to play a major role in the future energy system meeting the respective GHG reduction goals [1,2].

Especially hydrogen is seen as a promising secondary energy carrier due to its multifarious utilization possibilities. Particularly, hard to abate sectors like long-haul heavy-duty vehicles (HDV), aviation, shipping or steel production are promising applications for the use of hydrogen as energy source. As a result, a significant growing hydrogen demand is expected globally between 20 and 100 $Mt_{H2}$/a in 2030 and 90 to 700 $Mt_{H2}$/a in 2050 [3–7].

Furthermore, for countries with a high energy demand and limitations for sufficient domestic energy supply from renewables, the import of conditioned hydrogen as a "renewable" secondary energy carrier can help to cover the expected demand for carbon-neutral energy in the future. For Germany, the overall hydrogen demand is projected to increase to 5 to 24 $Mt_{H2}$/a in 2050. Thereof, imported hydrogen is expected to contribute between 5 and 9 $Mt_{H2}$/a (2050) to cover the projected demand. This results in a significant share of imported hydrogen at the overall hydrogen demand in Germany [8–12].

Since the hydrogen production cost, and therefore the cost of electricity from renewable sources, determines significantly the overall hydrogen supply cost, countries with a high solar radiation and/or high average wind velocities – in combination with large unused areas – are typically promising hydrogen exporting countries. However, the low volumetric energy density of hydrogen of around 3$kWh_{LHV}/m^3{}_{H2}$ [13] at ambient conditions does not allow the storage and transport of reasonable quantities taken economic constraints into consideration. Hence, hydrogen needs to be conditioned to increase its volumetric energy density. Broadly discussed options for this so-called conditioned hydrogen are physical conditioned compressed gaseous hydrogen (CGH2) as well as liquid hydrogen (LH2) and chemical conditioned hydrogen derivates like methanol (CH3OH), ammonia (NH3) or liquid organic hydrogen carrier (LOHC). Each option has its specific conditioning characteristics and different advantages and disadvantages relative to each other.

Thus the question needs to be answered which hydrogen supply pathway is the most cost-efficient green hydrogen supply option. Several studies carried out a techno-economic assessment of different hydrogen derivate supply chains. Nevertheless, most of the time, only two to three conditioned types of hydrogen have been considered, while the assumed input parameters vary for each study and therefore do not allow a direct comparison of the respective hydrogen supply cost. Additionally, often only an import or a national hydrogen production was taken into account and mostly neither a bandwidth, reflecting the high uncertainties in the underlying input parameters, nor more than one cost reference year (e.g., 2030 and 2050) was stated.

For example, Heuser et al. [14] did a techno-economic analysis of the hydrogen import based on water electrolysis powered by wind electricity from Patagonia to Japan only for a liquid hydrogen transport, calculating long term hydrogen supply cost of approximately 4.40 $€_{2020}/kg_{H2}$. Ratnakar et al. [15] identified great potential for liquid hydrogen to become an energy commodity like liquid natural gas. Giddey et al. [16] focused on ammonia for hydrogen supply and Timmerberg et al. [17] exclusively considered the hydrogen import from solar and wind energy from North Africa to Central Europe by a blend with natural gas to be transported within the existing natural gas pipelines leading to hydrogen supply cost to Central Europe between 1.80 and 4.00 $€_{2020}/kg_{H2}$. Al-Breiki et al. [18] made a cost comparison inter alia of liquid hydrogen, methanol and ammonia by just focusing on the energy carrier production, based on natural gas, and the maritime transportation cost identifying the overall lowest cost for liquid hydrogen and the highest for ammonia. Hong et al. [19] examined only the transport of hydrogen with the result that gaseous hydrogen transportation per pipeline has the lowest transportation cost from terminal to terminal for distances less than 2000 km compared to liquid hydrogen, LOHC and ammonia while Song et al. [20] identified LOHCs and liquid hydrogen as cheaper hydrogen supply options than gaseous hydrogen per pipeline for hydrogen filling stations in a case study for Shanghai. Teichmann et al. [21] assessed the energy consumption and the cost of the supply of selected LOHCs and liquid hydrogen per ship and truck with the result that LOHCs can be a viable option for hydrogen supply. Niermann et al. [22] additionally included compressed gaseous hydrogen and methanol as conditioned hydrogen as well as pipeline transportation showing that a gaseous hydrogen import per pipeline obtain the lowest overall hydrogen supply cost. However, both studies simplified the cost influence of the LOHC CAPEX, which is especially important for long transportation distances. Similar circumstances are given for the work of Hampp et al. [23] who did an techno-economic assessment of the hydrogen import cost to Germany from several different countries considering different conditioned hydrogen options and determined the lowest hydrogen supply cost to Germany in 2050 to 1.60 $€_{2020}/kg_{H2}$. Reuβ et al. [24] analyzed the cost of a hydrogen supply chain in 2050 in Germany from production up to the filling of full cell electric vehicles (FCEV) by taking compressed gaseous hydrogen, liquid hydrogen and a selected LOHC into account, while the import and further conditioned hydrogen options like methanol and ammonia have not been taken into consideration. They identified the transmission of gaseous hydrogen per pipeline to the filling station as the lowest cost supply options with around 6.10 $€_{2020}/kg_{H2}$ in 2050. Thereupon, Husarek et al. [25] considered also the import of hydrogen to Germany but only considered gaseous hydrogen for the national distribution and focused on the needed import quantities as well as pipeline capacities.

Against this background, this paper aims to assess the energy efficiency and cost of a green hydrogen supply from well to tank for hydrogen fueled HDV in Germany related to 2030 and 2050 by considering different hydrogen supply chains. To cover the high uncertainties of the techno-

economic parameters for the years 2030 and 2050, next to a baseline scenario, also a progressive and a conservative scenario are assessed indicating the overall bandwidth.

Hydrogen production is assumed by PEM water electrolysis based on electricity from wind turbines and photovoltaic systems. Since the electricity cost determine significantly the overall hydrogen supply cost, four different production regions are assumed.

- A local hydrogen production in closest proximity to the filling station in Central Germany with average solar radiation and wind velocities.
- A centralized hydrogen production in North Germany with a high offshore wind potential.
- A hydrogen production in North Africa (Tunisia) and Patagonia (Argentina) characterized by their outstanding high solar radiation respectively wind velocities.

The region-specific hydrogen production costs are determined by using a linear optimization model minimizing the cost by considering the regional wind velocities and solar radiation profiles as well as the annual costs of photovoltaic systems, wind turbines, potential electricity storage, electrolyzer, compressor and hydrogen storage.

For the transportation and storage of hydrogen from the place of production to the filling station, compressed gaseous hydrogen, liquid hydrogen, ammonia, methanol and liquid organic hydrogen carriers, namely dibenzyltoluene, are considered as conditioned hydrogen options. As transportation options, ships, pipelines and trucks are assumed. For the hydrogen filling, gaseous filling at 700 bar and additionally liquid hydrogen filling are taken into account since, for example, trucks driven by liquid hydrogen are under development [26]. Overall, this sums up to 34 different examined hydrogen supply chains.

## Supply chain definition

In this section, the assessed hydrogen supply chains are defined including the hydrogen production regions, the conditioned hydrogen options as well as the transportation and filling options.

### *Supply chain components*

Below the supply chain components are defined.

**Hydrogen Production Sites.** Hydrogen production typically contributes most to the overall hydrogen supply cost [22,24]. Thus, four different hydrogen production regions are defined displayed in Fig. 1, showing different representative characteristics regarding wind speed and solar radiation, availability of salt formations for a potential hydrogen storage in salt caverns as well as transportation distances to the end-consumer assumed located in Germany. A potential electricity storage using pumped stored hydroelectricity is not considered for all production sites.

- Local production Germany (LOC). The local decentralized hydrogen production represents a location for hydrogen production directly at the filling station; i.e. this is a decentralized supply option with no hydrogen transportation. Therefore, a spot in Central Germany is considered (51° 20′ N, 9° 30′ E) with average German renewable energy profiles. The full load hours are around 1100 h/a for photovoltaic systems and 2600 h/a for onshore wind power [27,28]. The abbreviation "LOC" is used for this production scenario.
- Centralized production Germany (GER). North Germany is assumed for the centralized German hydrogen production scenario due to its high offshore wind power potential and space availability. Since space constrains are given for onshore wind power, only photovoltaic (53° 60′ N, 7° 80′ E) and offshore wind (54° 50′ N, 5° 70′ E) power generation locations are assumed. The annual full load hours amount to ca. 1000 h/a for photovoltaic systems and up to 5000 h/a and more for offshore wind power [27,28]. The abbreviation "GER" describes this production case.
- Production Tunisia (TUN). Tunisia, classified as the only democracy in the whole Arab Region [29], is chosen as a hydrogen production country in North Africa due to its outstanding solar radiation. The selected location (33° 80′ N, 9° 90′ E) is characterized by annual full load hours of around 1800 h/a for photovoltaic systems and 4000 h/a for onshore wind turbines [27,28]. The import distance to Germany is rather short in a global context. The abbreviation "TUN" defines this region.
- Production Argentina (ARG). Hydrogen production in Argentina (Patagonia) is chosen due to its outstanding constant high wind velocities. Annual full load hours for the selected production region (52° 00′ S, 69° 00′ W) are stated to about 1000 h/a (photovoltaic) and up to 6000 h/a (onshore wind) [27,28,30]. However, the import distance to Germany is fairly long. "ARG" is the defined abbreviation.

In general, it should be pointed out that the average wind power annual full load hours, e.g., of the existing wind power park mix in Germany, are lower that the stated annual full load hours here [31]. This can be explained by the fact that the already existing and therefore older wind turbine types have a less advanced and adaptable design and hence significantly higher annual full load hours are expected for wind power in the future [32].

**Conditioned hydrogen options.** Due to hydrogens' low volumetric energy density, hydrogen needs to be conditioned to increase its volumetric energy density enabling economic storage and transportation. Promising options for the conditioned hydrogen considered here are the following [22,33–36]: (a) compressed gaseous hydrogen (CGH2), (b) liquid hydrogen (LH2) and the chemical bond of hydrogen to another molecule in the form of (c) methanol (CH3OH), (d) liquid ammonia (NH3) and (e) liquid organic hydrogen carrier (LOHC; exemplarily: dibenzyltoluene). Each option shows different energy densities (Fig. 2) and has its specific conditioning characteristics.

By compressing of gaseous hydrogen the volumetric energy density can be increased to more than 1 $MWh_{H2,LHV}/m^3_{CGH2}$. Alternatively hydrogen can be cooled down to −253 °C to be liquefied leading to a volumetric energy density of almost 2.4 $MWh_{H2,LHV}/m^3_{LH2}$ [13]. For both conditioning

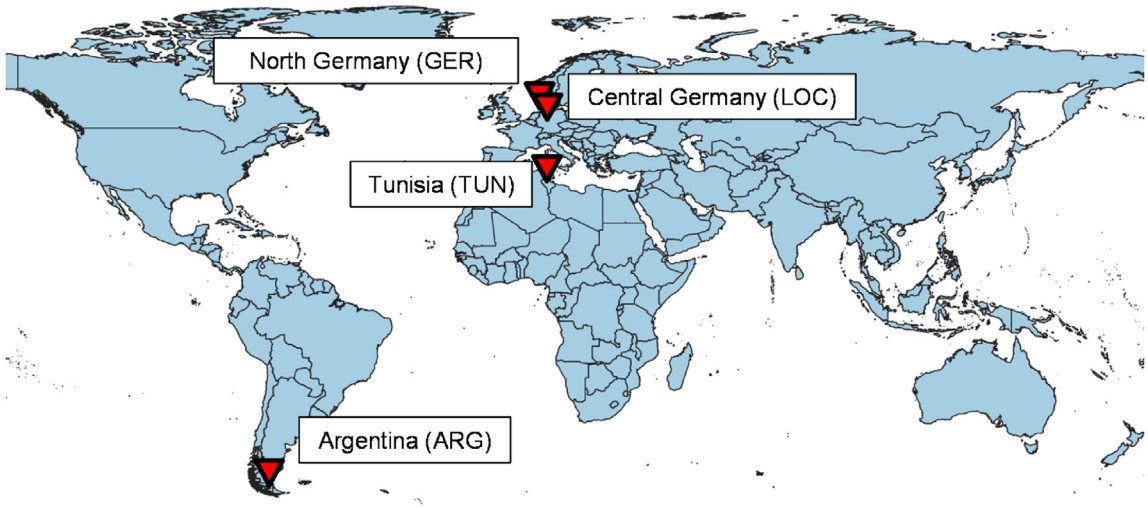


**Fig. 1 – Considered hydrogen production regions.**

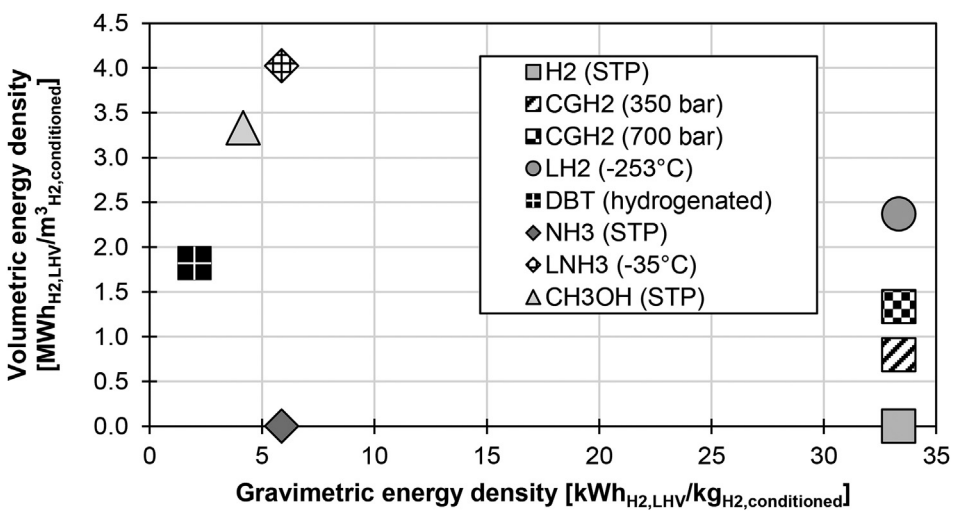


**Fig. 2 – Energy densities of conditioned hydrogen (values calculated based on [13,37,38]) (CGH2 = compressed gaseous hydrogen; CH3OH = methanol; DBT (LOHC) = Dibenzyltoluene; GH2 = gaseous hydrogen; NH3 = gaseous ammonia; LH2 = liquid hydrogen; LNH3 = liquid ammonia; STP = standard temperature and pressure).**

processes, electrical energy is needed. In addition, losses occur due to boil-off in the case of liquid hydrogen over time.

If hydrogen is chemically bound to another atom, the volumetric bounded hydrogen energy density can be even higher – but a carrier molecule is needed; e.g., for methanol, carbon dioxide ($CO_2$), and for ammonia, nitrogen ($N_2$), are needed as a hydrogen carrier. To release the pure hydrogen e.g. at the place of use and hence to crack the chemical bonds, high-temperature heat is needed (>300 °C). Additionally, a purification unit causing hydrogen losses. especially in the case of ammonia, is necessary to ensure the compulsory hydrogen quality for fuel cell applications. Since the hydrogen carrier are released into the atmosphere during this cracking process, there is the need to provide the carrier again at the production site. In the case of carbon dioxide, direct air capture and for nitrogen, air separation are assumed.

Similar circumstances are valid for the dehydrogenation of LOHCs. However, the LOHC is not released to the environment during dehydrogenation but is transported back to the hydrogenation location. Here, dibenzyltoluene (DBT) was considered as a carrier.

**Storage Options.** In the case of the liquid conditioned hydrogen options (liquid hydrogen, LOHC, methanol, ammonia) specific storage tanks are taken into account, e.g., for liquid hydrogen isolated cryo tanks or for methanol conventional mineral oil tanks. For compressed gaseous hydrogen, pressure tanks are assumed. Furthermore, salt caverns are assumed as a storage option after the hydrogen production in North Germany and Tunisa since salt formations are given [39]. Also for the seasonal storage of the central produced (GER) and imported compressed gaseous hydrogen (TUN) a salt cavern storage is assumed instead of a pressure tank.

**Transportation Options.** As import options, a ship transport of all liquid conditioned hydrogen options (liquid hydrogen, LOHC, methanol, ammonia) is assumed from the production area to Germany. In the case of compressed gaseous hydrogen import, pipeline transportation is considered. For the national distribution in Germany for all conditioned hydrogen options, road transportation per truck is assumed. In addition, the transport of compressed gaseous hydrogen in a pipeline network is taken into account.

**Filling Options.** A compressed gaseous hydrogen filling at 700 bar and a liquid hydrogen filling at −253 °C is considered. Hence, all conditioned hydrogen options need to be converted to gaseous respectively liquid hydrogen to be used by the HDV. The direct use of methanol (used by a methanol fuel cell), ammonia or LOHC are not considered as a fuel for the HDV since no announcement by OEMS for these concepts are known to the authors so far. The consequence is that chemical bound hydrogen has to be released from its carrier before the filling process. In contrast to the liquid hydrogen filling, where one technical solution is assumed, for the compressed gaseous hydrogen filling, three different filling station designs are considered depending on the type of hydrogen delivery.

- In the case of a compressed gaseous hydrogen delivery per truck, a swap trailer solution is assumed. Therefore, an additional moderate pressure tank is superfluous and only a high-pressure hydrogen tank (950 bar) is needed.
- If hydrogen is delivered per pipeline or by a local hydrogen release of the chemical hydrogen derivates, only a small, moderate pressure tank is needed since hydrogen supply is assumed to be almost constant.
- For a liquid hydrogen delivery per truck and compressed gaseous filling, the hydrogen can be stored as a liquid within a tank and being evaporated directly to high pressures (925 bar). Hence an evaporation unit, but no hydrogen compressor, is needed for this liquid hydrogen delivery and compressed gaseous hydrogen filling concept.

### *Hydrogen supply chains*

Based on these different components, 34 hydrogen supply chains are defined described below. A simplified visualization of these chains is given in Fig. 3 and Table 1 summarizes their specific characteristics. In the supplementary material, Figure A. 1 shows these hydrogen supply chains in detail with every single considered chain section.

The produced hydrogen is always converted in the production region to the specifically considered option of conditioned hydrogen. No further conversion to another conditioned hydrogen option is assumed during the supply chain to minimize the conversion steps and thus the conversion losses. Only at the supply chain end (i.e., the filling station), a reconversion to the needed hydrogen filling type (compressed gaseous hydrogen or liquid hydrogen) takes place.

For example, if ammonia is considered as a conditioned hydrogen option for the liquid hydrogen supply at the filling station, hydrogen is converted with nitrogen to ammonia in the production region. Subsequently, the hydrogen is transported and stored as ammonia with no further conversion steps up to the filling station location. Here, the ammonia is cracked and the released hydrogen is liquefied to be supplied in situ at the filling station for a subsequent filling into the HDV.

In addition, the assessment of the supply chains distinguishes between two different heat supply options for the reconversion processes of the liquid organic hydrogen carrier, methanol and ammonia. The first option is an external renewable heat supply. The second option considers that a fraction of the hydrogen delivered and released from the carrier is used internally to cover the given heat energy demand.

**Local Supply Chain (LOC).** The local supply chain is characterized by its simplicity since the hydrogen is produced directly at the filling station (i.e., no significant transportation is needed). Therefore, only compressed gaseous hydrogen respectively liquid hydrogen is considered for the local supply chain scenario.

Here, renewable electricity is generated with a photovoltaic and wind turbine system to be used subsequently as the energy input into the electrolyzer for hydrogen production. Thereupon, in the case of 700 bar filling, the gaseous hydrogen is compressed and stored in a pressure tank to cover given fluctuation and in the end, is filled at the filling station into the HDV.

For liquid hydrogen filling, the produced gaseous hydrogen is also compressed but stored in a smaller buffer pressure tank. Afterward, liquefaction takes place. Subsequently, the provided liquid hydrogen is pumped by cryo-pumps into a vacuum isolated cryo tank to harmonize production and demand. At the filling station this liquid hydrogen is filled into the HDV.

**National Supply Chain (GER).** Hydrogen is here assumed to be produced centralized in North Germany. Therefore, in the case of the pathways based on a liquid (liquid hydrogen, LOHC, methanol, ammonia), after hydrogen production, compression and buffer pressure tank storage, a short pipeline network is assumed to transport the hydrogen to the chain specific large-scale conversion plant. The potentially use of waste heat occurring during the conversion process is not taken into account since it is expected that in general no waste heat is locally needed at the hydrogen production sites. After the conversion, the resulting liquid is stored in the seasonal storage and distributed per truck to the filling stations. Here, the various conditioned hydrogen options are converted to the considered hydrogen filling option (compressed gaseous hydrogen or liquid hydrogen) and thereupon filled into the HDV. For the liquid hydrogen delivery to the filling station and a liquid filling, no additional conditioning is needed. In the case of a compressed gaseous hydrogen supply chain, two supply pathways are distinguished for each filling type. Both pathways consider a pipeline hydrogen transport to a consumption hub. From this hub, the first pathway defines the last mile hydrogen distribution per truck to the filling station, while the second pathway considers a direct pipeline link to the filling station.

**Import Supply Chain (TUN & ARG).** For the import supply chains, hydrogen production is identical to the other chains and all conditioned hydrogen options are again considered. However, it is assumed that the produced gaseous hydrogen

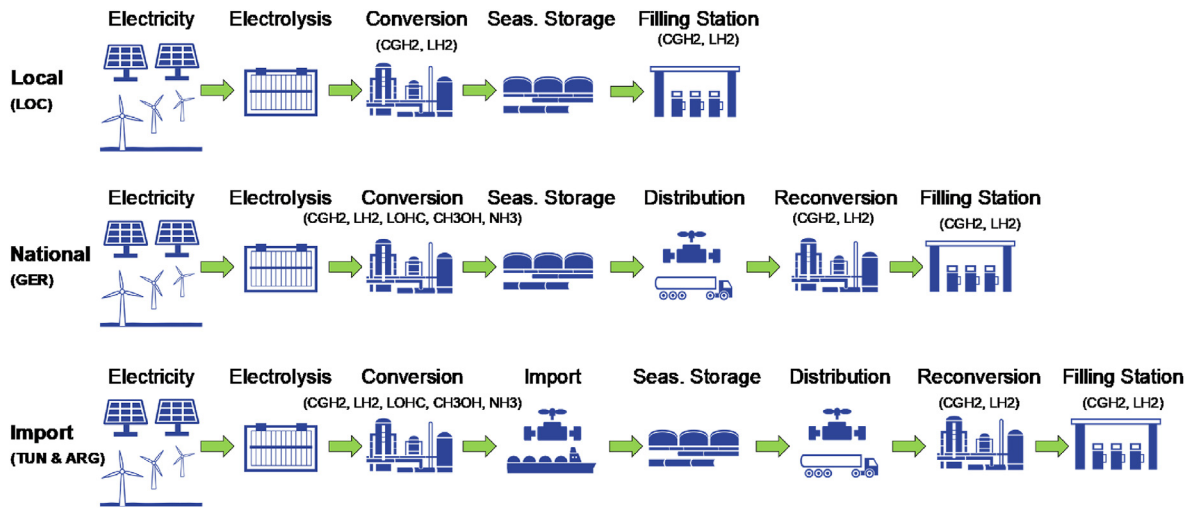


**Fig. 3 – Simplified visualization of assessed hydrogen supply chains (CGH2 = compressed gaseous hydrogen; CH3OH = methanol; GER = centralized production in North Germany; LH2 = liquid hydrogen; LOC = local production Central Germany; LOHC = liquid organic hydrogen carrier; NH3 = ammonia (liquid); Seas. = seasonal; TUN & ARG = production in Tunisia & Argentina (Patagonia)).**

**Table 1 – Main components of the assessed hydrogen supply chains (CH3OH = methanol; CGH2 = compressed gaseous hydrogen; CT = cryo tank; IT = isolated tank; LH2 = liquid hydrogen; LOHC = liquid organic hydrogen carrier; MOT = mineral oil tank; NH3 = ammonia; PT = pressure tank).**

| Supply chain | H2 prod. location | Cond. H2 option | Import | Distribution | H2 prod. s torage | Buffer storage | Seasonal storage |
|---|---|---|---|---|---|---|---|
| CGH2-LOC | Central Germany | CGH2 | – | – | - (PT[a]) | – | PT (CT[b]) |
| CGH2-Truck-Pip.-GER | North Germany | CGH2 | – | Truck/Pipeline[c] | – | PT | Cavern |
| CGH2-Pipe-Pip.-TUN | Tunisia | CGH2 | Pipeline | Truck/Pipeline[c] | – | PT | Cavern |
| CGH2-Pipe-Pip.-GER | North Germany | CGH2 | – | Pipeline | – | PT | Cavern |
| CGH2-Pipe-Pip.-TUN | Tunisia | CGH2 | Pipeline | Pipeline | – | PT | Cavern |
| LH2-Truck-GER | North Germany | LH2 | – | Truck | Cavern | CT | CT |
| LH2-Truck-Ship-TUN | Tunisia | LH2 | Ship | Truck | Cavern | CT | CT |
| LH2-Truck-Ship-ARG | Argentina | LH2 | Ship | Truck | PT | CT | CT |
| LOHC-Truck-GER | North Germany | LOHC | – | Truck | Cavern | MOT | MOT |
| LOHC-Truck-Ship-TUN | Tunisia | LOHC | Ship | Truck | Cavern | MOT | MOT |
| LOHC-Truck-Ship-ARG | Argentina | LOHC | Ship | Truck | PT | MOT | MOT |
| CH3OH-Truck-GER | North Germany | CH3OH | – | Truck | Cavern | MOT | MOT |
| CH3OH-Truck-Ship-TUN | Tunisia | CH3OH | Ship | Truck | Cavern | MOT | MOT |
| CH3OH-Truck-Ship-ARG | Argentina | CH3OH | Ship | Truck | PT | MOT | MOT |
| NH3-Truck-GER | North Germany | NH3 | – | Truck | Cavern | IT | IT |
| NH3-Truck-Ship-TUN | Tunisia | NH3 | Ship | Truck | Cavern | IT | IT |
| NH3-Truck-Ship-ARG | Argentina | NH3 | Ship | Truck | PT | IT | IT |

[a] If liquid hydrogen filling is assumed a PT is considered to enable a baseload liquefaction.
[b] If liquid hydrogen filling is assumed a CT is considered.
[c] The last 50 km are distributed per truck to the filling station from hydrogen hubs that are supplied by a national pipeline network; Cond. H2 = conditioned hydrogen, H2 prod. = hydrogen production.

is transported over a short distance per pipeline from the respective production region in Tunisia and Argentina (Patagonia) to the nearest export harbor, where the large-scale conversion of the specific conditioned hydrogen option takes place. Again, the potential use of the waste heat is not taken into account due to an expected lack of local demand. The needed water for the hydrogen production is produced by sea water desalination and transported per water pipeline to the location of the electrolyzer. Then the respective liquids are transported per overseas shipping to a harbor in North Germany. From here a subsequent transport to the seasonal storage facility is assumed to be realized per truck. In addition, for Tunisia also the import of compressed gaseous hydrogen per pipeline up to the seasonal storage is taken into account. Thereupon, the chain sections from the seasonal storage to the filling station are identical to the equivalent national supply chain variants for all import supply options.

## Methodology

Since the goal of this paper is a well to tank assessment of the energy efficiency and the cost of different conditioned hydrogen options for green hydrogen supply of fuel cell HDV, a technical as well as an economic assessment is realized. To do so, in a first step each specific needed single chain section is quantified regarding their techno-economic parameters and limitations for the in Chapter 2 defined supply chains. Then, these sections are treated as black boxes – regarding their needed capacities, mass and energy inputs/outputs to fulfil a normalized hydrogen demand at the filling station – and are connected with their resulting specifications to the next chain section, leading to the overall supply chain in the end. Since the hydrogen production is from major importance regarding the overall supply cost, this section is modelled in detail and will be shortly introduced in the following. Thereupon, the general assessment approach of the resulting overall supply chains will be presented.

### *Modelling approach*

To model the fluctuating and region-specific profiles of the solar radiation and wind velocities as a primary energy source adequately an further developed optimization model, based on [17] is used to estimate the region-specific hydrogen production cost. The designed model's objective is to identify system configurations with minimized hydrogen production cost for a defined hydrogen supply by taking all given constraints as well as the available system components (e.g. photovoltaic systems, wind turbines, electrolyzer, storage facility) into consideration.

This model uses a linear optimization approach and the objective function is shown in eq. (1). Thus the sum of all costs (*TC*) related to each component for the defined hydrogen supply is minimized. The parameter $TLCC_n$ is the total life cycle cost of component $n$ and includes the specific component cost. The specific costs are multiplied with the installed power, respectively capacity $W_n$ of component $n$.

$$TC = \min\left(\sum_{n=1}^{N} TLCC_n\, W_n\right) \tag{1}$$

To ensure that the given physical restrictions are considered, specific constraints are implemented. For example, the energy and mass flows must be balanced for every hour. Accordingly, balances for the electrical power and hydrogen mass flow are set up.

Eq. (2) determines the sum of the hourly electricity production ($Q_{el,production}$), where $f_{n,t}$ is the capacity factor of the renewable electricity generation technology $n$ [40] for each location at the time step $t$. Eq. (3) defines the hourly electricity supply ($Q_{el,supply}$) by considering the possible intermediate electricity storage battery ($Q_{el,bat,in}$ and $Q_{el,bat,out}$) and its specific storage efficiency ($\eta_{bat,out}$). Eq. (4) represents the hourly electricity demand ($Q_{el,demand}$): the sum of the demand from the electrolyzer ($Q_{el,pemel,in}$) and hydrogen compressor ($Q_{el,comp,in}$). As a critical additional constraint, eq. (5) ensures the covering of the hourly energy demand.

$$Q_{el,production} = \sum_{n=1}^{N} f_{n,t}\, W_n \tag{2}$$

$$Q_{el,supply} = Q_{el,production} + Q_{el,bat,out}\, \eta_{bat,out} - Q_{el,bat,in} \tag{3}$$

$$Q_{el,demand} = Q_{el,pemel,in} + Q_{el,comp,in} \tag{4}$$

$$Q_{el,demand} \leq Q_{el,supply} \tag{5}$$

Furthermore, the considered hydrogen storage technologies are limited by their capacities as well as input and output flows. Eq. (6) ensures that the state of charge of the hydrogen storage $n$ at the time step $t$ ($SOC_{H2,n,t}$) does not exceed the storage-specific installed capacity ($W_n$). The limitation of the hydrogen input and output flows of the hydrogen storage technology $n$ at the time step $t$ ($Q_{H2,n,in,t}$ and $Q_{H2,n,out,t}$) is formulated in eqs. (7) and (8), while eq. (9) determines the hydrogen charge level of the upcoming hour $t+1$ ($SOC_{H2,n,t+1}$).

$$SOC_{H2,n,t} \leq W_n \tag{6}$$

$$Q_{H2,n,out,t} \leq SOC_{H2,n,t} \tag{7}$$

$$Q_{H2,n,in,t} \leq W_n - SOC_{H2,n,t} \tag{8}$$

$$SOC_{H2,n,t+1} = SOC_{H2,n,t} + Q_{H2,n,in,t} - Q_{H2,n,out,t} \tag{9}$$

The estimation of the produced amount of hydrogen ($Q_{H2,pemel,out}$) takes place in eq. (10), where $Q_{el,pemel,in}$ describes the electricity input into the electrolyzer and $\eta_{pemel}$ describes the efficiency of the electrolyzer. Eq. (11) defines that the electrical input in the electrolyzer does not exceed its capacity ($Q_{W,pemel}$) determined by the installed power. Eq. (12) ensures that the hydrogen supply ($Q_{H2,supply}$) being the sum of the hydrogen output of the electrolyzer and the hydrogen storage ($Q_{H2,,,out}$) is always constant.

$$Q_{H2,pemel,out} = Q_{el,pemel,in}\, \eta_{pemel} \tag{10}$$

$$Q_{el,pemel,in} \leq Q_{W,pemel} \tag{11}$$

$$Q_{H2,supply} = Q_{H2,pemel,out} + Q_{H2,storage,out} = const. \tag{12}$$

The linear optimization takes place via the Python library PuLP. To reduce the computation time, an aggregation of typical occurring periods of renewable energy profiles is implemented. Therefore, a time series aggregation module [41] was used. It examines the time series for characteristic periods and aggregates them to condensed profiles.

### *Assessment approach*

**Technical Assessment.** The overall supply chain efficiency ($\eta_{SC}$) as the main technical assessment parameter analyzed here is calculated by using eq. (13). $q_{H2,nozzle}$ defines the energy content of the hydrogen at the nozzle of the filling station and $q_{overall}$ is the overall energy input for the hydrogen supply from well to nozzle. The energy content of the hydrogen always refers to the lower heating value (LHV).

$$\eta_{SC} = q_{H2,nozzle} \big/ q_{overall} \tag{13}$$

The overall energy input $q_{overall}$ is defined as the sum of energy inputs for every chain section (eq. (14)) (i.e., hydrogen production ($q_{production}$), hydrogen conversion to the specific conditioned hydrogen option ($q_{conversion}$), storage ($q_{storage}$), transport ($q_{transport}$), reconversion ($q_{reconversion}$), hydrogen filling ($q_{fill}$)).

$$q_{overall} = q_{production} + q_{conversion} + q_{storage} + q_{transport} + q_{reconversion} + q_{fill} \tag{14}$$

Hydrogen losses throughout the overall supply chain (e.g., caused by boil-off during storage and transportation or purification losses in the reconversion) are compensated by an additional hydrogen production. The respective energy input is calculated with eq. (15). It distinguishes between the energy input for the production of one unit of hydrogen at the production site ($q_{production,ideal}$) and the additional energy input caused by hydrogen losses ($q_{production,losses}$) to supply one unit of hydrogen at the filling stations nozzle.

$$q_{production} = q_{production,ideal} + q_{production,losses} \tag{15}$$

For the hydrogen compression energy input, being part of the energy input for the conversion section ($q_{conversion}$), the electric energy demand strongly depends on the inlet and outlet pressure. Therefore, the necessary energy is calculated based on the specific inlet and outlet pressure.

In the case of ionic compressors, quasi-isothermal compression can be assumed [42]. Hence an isothermal compression of a real gas by considering the compressibility factor of hydrogen is used for the calculation of the compression energy $q_{comp}$ (eq. (16)) [31]. $R_s$ is the specific gas constant of hydrogen, $\eta_{Comp}$ the compressor efficiency including the isothermal, mechanic and electric efficiency of the compressor, $T_{comp}$ the compression temperature, and $z_{out}$ and $z_{in}$ the compressibility factor of hydrogen at the outlet ($p_{out}$) and inlet pressure ($p_{in}$) of the hydrogen.

$$q_{comp} = \frac{R_s\, T_{comp}}{\eta_{comp}} \left[ z_{out} \ln\left(\frac{p_{out}}{1.0135\ bar}\right) - z_{in} \ln\left(\frac{p_{in}}{1.0135\ bar}\right) \right] \tag{16}$$

The compressibility factors are estimated by eq. (17) with the parameters $B_i$ and $C_i$ derived from eq. (18) and (19) using the coefficients $a_i$, $b_i$ and $c_i$ listed in T. 1 in the appendix [43].

$$z_i = 1 + \frac{B_i}{R_s\, T_{comp}}\, p_i + \frac{C_i - B_i^{\,2}}{\left(R_s\, T_{comp}\right)^2}\, p_i^{\,2} \tag{17}$$

$$B_i = a_1\, T_{comp}^{\;2} + b_1\, T_{comp} + c_1 \tag{18}$$

$$C_i = a_2\, T_{comp}^{\;2} + b_2\, T_{comp} + c_2 \tag{19}$$

**Economic Assessment.** The overall hydrogen supply cost ($c_{overall}$) are used for a quantitative comparison of the different hydrogen supply chains. Therefore, first all economic parameters are inflation-adjusted to euros (€$_{2020}$) for the year 2020 [44,45].

The overall hydrogen supply costs are, similar to the calculation of the overall energy inputs, the sum of the chain section costs represented by the costs of hydrogen production ($c_{production}$), conversion ($c_{conversion}$), storage ($c_{storage}$), transport ($c_{transport}$) and reconversion ($c_{reconverison}$) as well as for the hydrogen filling ($c_{fill}$) (eq. (20)).

$$c_{overall} = c_{production} + c_{conversion} + c_{storage} + c_{transport} + c_{reconversion} + c_{fill} \tag{20}$$

The specific chain section costs ($c_{section,i}$) are the sum of the annuity of the specific chain sections $i$ capital expenditures ($ACAPEX_i$) and the annual operational expenditures ($OPEX_i$) including energy-related expenses (e.g., the dehydrogenation heat demand for the LOHC) divided by the annual filled hydrogen mass ($m_{H2,fill,annual}$) (eq. (21)).

$$c_{section,i} = \frac{ACAPEX_i + OPEX_i}{m_{H2,fill,annual}} \tag{21}$$

The annual mass of filled hydrogen is the annual mass of produced hydrogen ($m_{H2,prod,annual}$) minus the annual mass of hydrogen losses ($m_{H2,loss,annual}$) mainly due to among others boil-off and purification.

$$m_{H2,fill,annual} = m_{H2,prod,annual} - m_{H2,loss,annual} \tag{22}$$

The annuity of the specific chain sections capital expenditures is calculated based on the annuity method by considering technology-specific depreciation time periods ($d_i$) – always based on the technology-specific technical lifetime –, the real weighted average cost of capital ($WACC_{real}$) and the technology-specific capital expenditures ($CAPEX_i$) (eq. (23)) [46].

$$ACAPEX_i = CAPEX_i \frac{WACC_{real}\,(1 + WACC_{real})^{d_i}}{(1 + WACC_{real})^{d_i} - 1} \tag{23}$$

The real weighted average cost of capital is calculated to 6% by using eq. (24), with the nominal weighted average cost of capital ($WACC_{nom}$) assumed to 8% and a future inflation rate ($INFL$) of around 2% based on the targets of the ECB and FED [47,48].

$$WACC_{real} = \frac{1 + WACC_{nom}}{1 + INFL} - 1 \tag{24}$$

Especially for the conversion and reconversion processes, the specific capital expenditures vary strongly depending on the considered scale. Thus, a scaling function is assumed (eq. (25)) [49] with $CAPEX_{i,design}$ as the technology-specific capital expenditure of the designed plant size ($S_{i,design}$), $CAPEX_{i,base}$ as the technology-specific capital expenditure for the base plant size ($S_{i,base}$) and $\alpha_i$ as the technology-specific investment scaling factor.

$$CAPEX_{i,design} = CAPEX_{i,base} \left(\frac{S_{i,design}}{S_{i,base}}\right)^{\alpha_i} \tag{25}$$

For some technologies (e.g., storage systems), the capital expenditures do not include the installation factor ($f_{inst}$). In this case, the installation cost is considered by using eq. (26) to determine the technology-specific installed capital expenditures ($CAPEX_{i,inst}$).

$$CAPEX_{i,inst} = CAPEX_{i,design}\, f_{inst} \tag{26}$$

Unlike to ammonia and methanol, the carrier molecule of the liquid organic hydrogen carrier is not released to the environment after dehydrogenation. Typically, a circulation

**Table 2 – Frame conditions for the different production regions and resulting supply chains for the hydrogen production local at the filling station (LOC), centralized in North Germany (GER), in Tunisia (TUN) and in Argentina (ARG).**

| Parameter | Unit | Year | LOC [50,51,55] | GER [50–55] | TUN [51–56] | ARG [51–56] |
|---|---|---|---|---|---|---|
| Seasonal storage capacity | days | both | 15 | 30 | 60 | 60 |
| Import distance | km | both | – | – | 5,000[a] (3,000[b]) | 14,400[a] |
| Distribution distance[c] | km | both | – | 350 | 350 | 350 |
| Baseload electricity cost | €$_{2020}$/MWh$_{el}$ | 2030 | 80 (65–95) | 80 (65–95) | 60 (50–70) | 50 (40–60) |
| | | 2050 | 60 (50–70) | 60 (50–70) | 40 (30–50) | 35 (30–40) |
| Baseload heat cost | €$_{2020}$/MWh$_{el}$ | 2030 | 80 (65–95) | 80 (65–95) | 60 (50–70) | 50 (40–60) |
| | | 2050 | 60 (50–70) | 60 (50–70) | 40 (30–50) | 35 (30–40) |

[a] For ship import.
[b] For pipeline import.
[c] Total average transportation distance in Germany from production respectively import harbor to seasonal storage and consequently to the filling station (e.g. Hamburg to Ruhr-Area).

**Table 3 – Techno-economic parameters for photovoltaic, onshore and offshore wind power systems (numbers in brackets define the progressive (lower bandwidth) and conservative values (upper bandwidth); all values are derived from the median (baseline), first and third quartile (progressive and conservative)).**

| Parameter | Unit | Year | Photovoltaic [1,57–60] | Onshore Wind [1,60–63] | Offshore Wind [1,60–63] |
|---|---|---|---|---|---|
| CAPEX | €$_{2020}$/kW$_{el}$ | 2030 | 400 (310–570) | 1110 (1010–1240) | 1890 (1750–2020) |
| | | 2050 | 270 (170–350) | 990 (860–1140) | 1620 (1320–1930) |
| OPEX | %$_{CAPEX}$/a | 2030 | 2.6 | 2.5 | 2.8 |
| | | 2050 | 2.8 | 2.3 | 2.5 |
| Depreciation | a | 2030 | 30 | 30 | 30 |
| | | 2050 | 30 | 30 | 30 |

within the supply chain is realized. Therefore, the specific annual cost for the liquid organic hydrogen carrier per unit released hydrogen ($c_{LOHC}$) has to be taken into consideration and is calculated based on eq. (27). $CAPEX_{LOHC}$ is the capital expenditure for the liquid organic hydrogen carrier per unit hydrogenated hydrogen capacity. $Cycle_{LOHC,annual}$ are the annual achievable round trip cycles of the liquid organic hydrogen carrier, while one round trip cycle is defined as the hydrogenation (conversion) of the liquid organic hydrogen carrier at the production site, the storage, transportation and dehydrogenation (reconversion) at the consumption site and re-transportation to the production site. Additionally, since repeated hydrogenation-cycles lead to a deterioration of the liquid organic hydrogen carrier, the depreciation time of the LOHC ($d_{LOHC}$) is taken into account and set equivalent to its technical lifetime.

$$c_{LOHC} = \frac{CAPEX_{LOHC}}{Cycle_{LOHC,annual}} \frac{WACC_{real}\,(1+WACC_{real})^{d_{LOHC}}}{(1+WACC_{real})^{d_{LOHC}}-1} \tag{27}$$

The annual achievable round trip cycles of the liquid organic hydrogen carrier are estimated with eq. (28) where $t_{LOHC,\ cycle}$ is the time per liquid organic hydrogen carrier round trip cycle.

$$cycle_{LOHC,annual} = 8760\frac{h}{a}\Big/ t_{LOHC,cycle} \tag{28}$$

The time per round trip cycle is calculated with eq. (29) by the sum of the average dwell times of the liquid organic hydrogen carrier in the chain sections conversion ($t_{LOHC,conversion}$), storage ($t_{LOHC,storage}$), transport ($t_{LOHC,transport}$) and reconversion ($t_{LOHC,reconverison}$).

$$t_{LOHC,cycle} = t_{LOHC,conversion} + t_{LOHC,storage} + t_{LOHC,transport} + t_{LOHC,reconversion} \tag{29}$$

## Data, assumptions and framework

This section presents the input data for the assessment, considering a baseline, progressive and conservative scenario. Additionally, for technologies with significant expected cost reductions or technology improvments, crucial parameters are differentiated between the years 2030 and 2050.

**Table 4 – Techno-economic parameters for the Li-Ion battery (numbers in brackets define the progressive (lower bandwidth) and conservative values (upper bandwidth)).**

| Parameter | Unit | Year | Li-Ion battery [9,58,59,63,64] |
|---|---|---|---|
| CAPEX | €$_{2020}$/kWh$_{el}$ | 2030 | 145 (118–228) |
| | | 2050 | 88 (70–98) |
| OPEX | %$_{CAPEX}$/a | 2030 | 2 |
| | | 2050 | 4 |
| Depreciation | a | both | 15 |
| Efficiency[a] | % | both | 90 |

### Frame conditions

Depending on the hydrogen production region, different characteristics regarding seasonal storage capacities, import distances and transportation distances in Germany from the seasonal storage to the filling station are taken into account (Table 2). Additionally, the costs for a baseload supply of heat and electricity based on renewable energies, considered for the hydrogen conditioning processes and varying per region, need to be defined. The cost for heat – needed at a temperature level above 300 °C for hydrogen release from the carriers – is based on the electricity cost. The seasonal storage capacity for a local hydrogen production directly at the filling station (LOC) is based on the duration of dark dulls described in Refs. [50,51], while the seasonal storage capacity for the centralized hydrogen production in North Germany (GER) is set to 30 days to consider in addition to the dark doldrums a strategic capacity. In comparison to this, the seasonal storage capacities of 60 days for the hydrogen importing scenarios (TUN & PAT) is higher due to a higher assumed share of strategic reserve and therefore independence from other countries and in the range of today's capacities for crude oil and natural gas strategic reserves [52–54].

### Hydrogen production

Green hydrogen is produced by water electrolysis with an electrolyzer operated by green electricity and water. The assumed parameters for electricity generation by photovoltaic systems and wind turbines are listed in Table 3. Table 4 shows the assumed parameters for the electricity storage by Li-Ion batteries and Table 5 the considered data for the PEM-electrolyzer. In the case of the electrolyzer the techno-economic parameters are based on large-scale systems (100 MW+) taken already the specific cost decline due to scaling effects into account. All costs are given as overall costs including all auxiliaries, installation etc.

For photovoltaic, wind power and PEM-electrolyzer, the baseline capital expenditures (CAPEX) are derived from the median of the data input, while the first and third quartile is used for the progressive and conservative scenarios. The water supply for the PEM-electrolyzer is assumed by reverse osmose seawater desalination and consecutive pipeline transport to the PEM-electrolyzer location for Tunisia and Argentina (Patagonia). For Germany, the water supply cost are based on a given water pipeline infrastructure.

### Conditioning

The parameters for the gaseous hydrogen compression, for hydrogen liquefaction and liquid hydrogen pumps as well as for the conversion and reconversion of the liquid organic hydrogen carrier, methanol and ammonia are listed Table 6. Again, the bandwidths for highly cost-sensitive parameters as capital expenditures (CAPEX) or hydrogen losses are considered.

### Storage

In order to balance the fluctuating renewable energy production relative to the hydrogen demand, hydrogen respectively the conditioned hydrogen need to be stored. Therefore, Table 7 shows the assumptions of the techno-economic parameters of the conditioned hydrogen options storage systems. All parameters are based on large-scale systems.

**Table 5 – Techno-economic parameters of the PEM-electrolyzer and water supply by reverse osmoses desalination and pipeline transport (numbers in brackets define the progressive (lower bandwidth) and conservative values (upper bandwidth); values for CAPEX, OPEX, depreciation and efficiency derived from the median (baseline), first and third quartile (progressive and conservative)).**

| Parameter | Unit | Year | PEM-electrolyzer [62,64–82] |
|---|---|---|---|
| CAPEX | $€_{2020}/kW_{el}$ | 2030 | 860 (580–1230) |
| | | 2050 | 510 (350–760) |
| OPEX | $\%_{CAPEX}/a$ | 2030 | 3.5 |
| | | 2050 | 2.8 |
| Depreciation | a | 2030 | 20 |
| | | 2050 | 25 |
| Efficiency[a] | % | 2030 | 67 (63–69) |
| | | 2050 | 71 (67–74) |
| $H_2O$ demand | $kg_{H2O}/kg_{H2}$ | both | 13 |
| $H_2O$ supply cost[b] | $€_{2020}/m^3_{H2O}$ | both | 2.11 |
| $H_2O$ desalination cost | $€_{2020}/m^3_{H2O}$ | both | 2.62 |
| $H_2O$ desalination electricity demand | $kWh_{el}/m^3_{H2O}$ | both | 3.1 |
| $H_2O$ transportation cost[c] | $(€_{2020}/m^3_{H2O})/100$ km | both | 0.55 |
| $H_2O$ transportation electricity demand[c] | $(kWh_{el}/m^3_{H2O})/100$ km | both | 0.45 |

[a] System efficiency based on the LHV ($kWh_{H2.\ LHV}/kWh_{el,\ input}$).
[b] For a given water pipeline infrastructure.
[c] Based on pipeline transportation.

**Table 6 – Techno-economic parameters of the gaseous hydrogen compressor (CGH2), hydrogen liquefier (LH2) and liquid hydrogen pump (LH2 pump) as well as of the conditioning of liquid organic hydrogen carriers (LOHC), methanol (CH3OH) and ammonia (NH3) (numbers in brackets define the progressive (lower bandwidth) and conservative values (upper bandwidth); values are valid for 2030 and 2050).**

| Parameter | Unit | CGH2 [24,42,43,49,83–85] | LH2 [24,49,83–87] | LH2 pump [64,83,84] |
|---|---|---|---|---|
| $CAPEX_{base}$ | $M€_{2020}$ | 0.56 (0.37–1.16) | 195 (139–235) | 0.16 (0.13–0.18) |
| $Scale_{base}$ | $t_{H2}/d$ | 6 | 100 | 6 |
| $Scale_{design}$ | $t_{H2}/d$ | 6 (3[a]) | 100 (3[a]) | 6 (3[a]) |
| Installation factor | – | 1.25 | 1[b] | 1.25 |
| Scaling factor | – | 0.84 | 0.77 | 1 |
| OPEX | $\%_{CAPEX}/a$ | 5 | 4 | 3 |
| Depreciation | a | 15 | 30 | 10 |
| Electricity demand | $kWh_{el}/kg_{H2}$ | calc. | 7.4 (6.0–9.8) | 0.1 |
| Compressor efficiency | % | 79 (68–85) | – | – |
| $H_2$ loss[c] | % | 0.5 (0.0–1.0) | 1.5 (1.0–2.0) | 0.7 (0.0–1.3) |
| **Parameter** | **Unit** | **LOHC[22,24,37,49,64,88–91]** | **CH3OH [22,49,92–103]** | **NH3 [16,38,49,64,89,90,104–111]** |
| **Conversion** | | | | |
| $CAPEX_{base}$ | $M€_{2020}$ | 23.4 (19.7–24.1) | 65.3 (44.2–76.8) | 109 (93.4–125) |
| $Scale_{base}$ | $t_{H2}/d$ | 100 | 100 | 100 |
| $Scale_{design}$ | $t_{H2}/d$ | 100 (3[a]) | 100 (3[a]) | 100 (3[a]) |
| Installation factor | – | 1[b] | 1[b] | 1[b] |
| Scaling factor | – | 0.65 | 0.85 | 0.85 |
| OPEX | $\%_{CAPEX}/a$ | 4 | 4 | 4 |
| Carrier cost | $€_{2020}/kg_{carrier}$ | 2.4 (1.7–4.1) | 0.15 (0.08–0.3) | 0.03 (0.02–0.04) |
| Carrier energy demand | $kWh/kg_{carrier}$ | 0 | 1.7 (0.9–2.1) | 0.11 (0.09–0.13) |
| Carrier H2 capacity | wt% | 5.6[d] | 17.65 | 12.5 |
| Depreciation | a | 20 (15[e]) | 20 | 20 |
| Electricity demand | $kWh_{el}/kg_{H2}$ | 0.3 | 0.7 | 1.8 (1.4–2.3) |
| Heat demand | $kWh_{th}/kg_{H2}$ | −9.9 | −0.4 | −6.4 |
| $H_2$ loss[c] | % | 0.5 (0.1–1.0) | 33[f] | 0.5 (0.1–1.0) |
| **Reconversion** | | | | |
| $CAPEX_{base}$ | $M€_{2020}$ | 20.3 (14.8–26.0) | 21.1 (15.4–44.1) | 63.7 (47.9–105) |
| $Scale_{base}$ | $t_{H2}/d$ | 100 | 100 | 100 |
| $Scale_{design}$ | $t_{H2}/d$ | 3[a] | 3[a] | 3[a] |
| Installation factor | – | 1[b] | 1[b] | 1[b] |
| Scaling factor | – | 0.65 | 0.85 | 0.85 |
| OPEX | $\%_{CAPEX}/a$ | 4 | 4 | 4 |
| Depreciation | a | 20 | 20 | 20 |
| Electricity demand | $kWh_{el}/kg_{H2}$ | 1.1 (0.7–1.5) | 4.2 (3.9–4.5) | 3.4 (2.8–4.0) |
| Heat demand | $kWh_{th}/kg_{H2}$ | 11.2 (11.0–12.0) | 3.6 (3.2–4.4)[g] | 9.7 (9.0–10.1)[g] |
| $H_2$ loss[c] | % | 2 (1–3) | −28 (−35 to −20)[h] | 20 (10–30) |

[a] In the case of decentral conditioning.
[b] Already included in $CAPEX_{base}$.
[c] Hydrogen loss per unit hydrogen conditioned.
[d] Dibenzyltoluene can hold up to 6.2 wt% $H_2$ [37], but a maximum depth-of-discharge for cycling at 90% is advisable for (de-)hydrogenation [91].
[e] For the carrier (dibenzyltoluene).
[f] One-third of the hydrogen reacts to water.
[g] Hydrogen losses can be used to cover its equivalent of heat demand.
[h] Additional to the released hydrogen stored in the methanol a 50% surplus of hydrogen is gained from the water splitting, while in the purification process 10–20% hydrogen losses occur.

### *Transportation*

Since renewable hydrogen production and consumption are likely to be spatially decoupled, hydrogen needs to be transported. Thus Table 8 shows the techno-economic parameters for road transportation. For the tractor, a fuel cell truck powered by hydrogen is assumed to enable a climate-neutral transport. Additionally, Table 9 shows the assumed parameters for the conditioned hydrogen import by ship. In the case of the liquid hydrogen ship, the boil-off losses are assumed to be used as a fuel, while for the other ships, a PtL fuel is assumed. Adding up to this, Table 10 contains the considered techno-economic parameters for the gaseous hydrogen transport per pipeline.

- A pipeline with a diameter of 1400 mm can transport large volumes of hydrogen and is representative for an import pipeline.
- The 400 mm diameter pipeline is considered for the national transport.
- The 100 mm diameter pipeline for the last mile distribution (e.g., directly connected to the filling station).

**Table 7 – Techno-economic parameters of the gaseous hydrogen pressure tank (CGH2 PT), cavern (CGH2 cavern) and liquid hydrogen (LH2) storage (numbers in brackets define the progressive (lower bandwidth) and conservative values (upper bandwidth); values are valid for 2030 and 2050).**

| Parameter | Unit | CGH2 PT [24,83–85,89,95,112,113] | CGH2 cavern [83,114,115] | LH2 [24,64,84,85,89,116] |
|---|---|---|---|---|
| CAPEX | €$_{2020}$/kg$_{H2}$ | 460 (220–600) | 46 (12–128) | 33 (14–105) |
| Inst. factor | – | 1.3 | 1[a] | 1.25 |
| OPEX | %$_{CAPEX}$/a | 2 | 2 | 2 |
| Depreciation | a | 30 | 30 | 30 |
| Pressure[b] | bar | 100 | 100 | 1 |
| $H_2$ loss[c] | %/d | 0 | 0 | 0.1[d] |
| **Parameter** | **Unit** | **LOHC [64,84,89,116]** | **CH3OH [64,84,89,116]** | **NH3 [64,84,89,109,117]** |
| CAPEX | €$_{2020}$/kg$_{H2}$ | 18 (13–30) | 4 (3–6) | 13 (5–15) |
| Inst. factor | – | 1.3 | 1.3 | 1.3 |
| OPEX | %$_{CAPEX}$/a | 2 | 2 | 2 |
| Depreciation | a | 30 | 30 | 30 |
| Pressure[b] | bar | 1 | 1 | 1 |
| $H_2$ loss[c] | %/d | 0 | 0 | 0.03 |

[a] Already included into CAPEX.
[b] Average storage pressure level.
[c] Loss per day per unit stored hydrogen.
[d] Based on large scale liquid hydrogen storage, in the case of small scale storage systems (e.g., vehicle tanks), losses can be significantly higher due to poor surface to volume ratio.

**Table 8 – Techno-economic parameters of the fuel cell truck (FC truck), compressed gaseous hydrogen, liquid hydrogen, liquid organic hydrogen carrier (LOHC), methanol (CH3OH) and liquid ammonia (NH3) trailer (numbers in brackets define the progressive (lower bandwidth) and conservative values (upper bandwidth); values are valid for 2030 and 2050).**

| Parameter | Unit | FC tractor [116,118–120] | CGH2 [22,64,85,116,121] | LH2 [22,64,85,89,116,121] | LOHC [22,64,116,121] | CH3OH [22,64,116,121] | NH3 [64] |
|---|---|---|---|---|---|---|---|
| CAPEX | k€$_{2020}$/unit | 190 (140–270) | 680 (520–1060) | 910 (740–1530) | 110 | 110 | 200 |
| OPEX | %$_{CAPEX}$/a | 12 | 2 | 2 | 2 | 2 | 2 |
| Depr. | a | 8 | 12 | 12 | 12 | 12 | 12 |
| Oper. cost | €$_{2020}$/h | 36 | – | – | – | – | – |
| Utilization | h/a | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 |
| Velocity | km/h | 50 | – | – | – | – | – |
| Fuel dem. | kWh$_{H2}$/km | 2.3 | – | – | – | – | – |
| Fuel cost | €$_{2020}$/kWh$_{H2}$ | 0.18 | – | – | – | – | – |
| Payload | kg$_{H2}$/unit | – | 1100 | 4300 | 1670 | 2930 | 2600 |
| Lo. time | h/load | – | 1.5 | 3 | 1.5 | 1.5 | 1.5 |
| Pressure[a] | bar | – | 500 | 1 | 1 | 1 | 1 |
| $H_2$ loss | %/d | – | 0 | 0.15[b] (0.0–0.3) | 0 | 0 | 0 |

[a] Storage pressure level of a full trailer.
[b] Depends strongly on transportation time, up to one day of storage with no boil-off losses due to capable pressure increase; Depr. = depreciation; Lo. = loading/unloading; Oper. = operating (driver) cost.

*Filling stations*

The efficiency and cost for the hydrogen filling stations strongly depend on the type of the hydrogen delivery concept due to the specific design (e.g., buffer storage or compressor capacity can vary tremendously). Due to the delivery concept-specific design variations and the novelty of liquid hydrogen filling stations no public data was available at least to the authors. Therefore, own assumptions regarding the capacities of the storage unit as well as of the power of the compressor,

**Table 9 – Techno-economic parameters of the liquid hydrogen (LH2), liquid organic hydrogen carrier (LOHC), methanol (CH3OH) and liquid ammonia (NH3) ships (numbers in brackets define the progressive (lower bandwidth) and conservative values (upper bandwidth); values are valid for 2030 and 2050).**

| Parameter | Unit | LH2 [14,21,22,64,122] | LOHC [21,64,123] | CH3OH [21,64,123] | NH3 [18,21,64,117] |
|---|---|---|---|---|---|
| CAPEX | $M€_{2020}$/ship | 387 (195–478) | 83 (70–96) | 83 (70–96) | 79 (72–83) |
| OPEX | $\%_{CAPEX}$/a | 4 | 4 | 4 | 4 |
| Depreciation | a | 25 | 25 | 25 | 25 |
| Operating cost | $€_{2020}$/h | 572 | 572 | 572 | 572 |
| Utilization | h/a | 8000 | 8000 | 8000 | 8000 |
| Velocity | km/h | 33 | 30 | 30 | 30 |
| Fuel demand | $kWh_{PtX}$/km | 0[a] | 920 | 920 | 690 |
| Fuel cost | $€_{2020}/kWh_{PtX}$ | -[a] | 0.16 | 0.16 | 0.16 |
| Payload | $t_{H2}$/ship | 11,000 | 6600 | 13,750 | 9350 |
| De/Loading time[b] | h/load | 54 | 54 | 54 | 54 |
| $H_2$ loss | %/d | 0.2 (0.15–0.3) | 0 | 0 | 0 |
| $H_2$ flash[c] | %/load | 1.3 (0.9–1.7) | – | – | – |

[a] Boil-off losses assumed as fuel and therefore no additional fuel demand.
[b] Includes waiting time at the port.
[c] Additional hydrogen losses during loading.

**Table 10 – Techno-economic parameters of the compressed gaseous hydrogen pipelines (numbers in brackets define the progressive (lower bandwidth) and conservative values (upper bandwidth); values are valid for 2030 and 2050).**

| Parameter | Unit | 1400 mm [64,113,124] | 400 mm [64,113,124] | 100 mm [64,113,124] |
|---|---|---|---|---|
| CAPEX | $M€_{2020}$/km | 6.58 (4.75–10.3) | 1.08 (0.85–1.38) | 0.41 (0.34–0.49) |
| OPEX | $\%_{CAPEX}$/a | 5 | 5 | 5 |
| Depreciation | a | 40 | 40 | 40 |
| Pressure | bar | 100 | 65 | 30 |
| Capacity factor | % | 90 | 90 | 90 |
| $H_2$ transport | $kt_{H2}$/a | 5190 | 278 | 8 |
| Electricity demand | $kWh_{el}/(kg_{H2}$ 100 km) | 0.06 (0.04–0.1) | 0.06 (0.04–0.1) | 0.06 (0.04–0.1) |
| $H_2$ loss | $kg_{H2}/(kg_{H2}$ 100 km) | 0 | 0 | 0 |

**Table 11 – Techno-economic parameters of the compressed gaseous hydrogen filling station considering a compressed gaseous hydrogen trailer swap delivery (CGH2 swap), pipeline respectively local hydrogen derivate reconversion delivery (CGH2 pipe/recon) or liquid hydrogen delivery per trailer and evaporation (CGH2 liquid) and techno-economic parameters for the liquid hydrogen filling station (LH2) assuming a liquid hydrogen delivery per trailer or local liquefaction (all values are based on own design assumptions, derived from the component specifications in Ref. [85]; numbers in brackets define the progressive (lower bandwidth) and conservative values (upper bandwidth)).**

| Parameter | Unit | Year | CGH2 swap | CGH2 pipe/recon | CGH2 liquid | LH2 |
|---|---|---|---|---|---|---|
| CAPEX | $M€_{2020}$/unit | 2030 | 5.0 (4.0–6.3) | 4.4 (3.5–5.5) | 4.6 (3.7–5.8) | 3.6 (1.8–7.2) |
| | | 2050 | 4.0 (3.2–5.0) | 3.4 (2.7–4.3) | 2.5 (2.0–3.1) | 2.5 (1.3–5.0) |
| OPEX | $\%_{CAPEX}$/a | both | 5 | 5 | 5 | 5 |
| Filling capacity | $kg_{H2,max}$/h | both | 320 | 320 | 320 | 320 |
| | $kg_{H2,max}$/d | both | 4990 | 4990 | 4990 | 7680 |
| Depreciation | a | both | 15 | 15 | 15 | 15 |
| Capacity factor[a] | % | both | 50 (40–60) | 50 (40–60) | 50 (40–60) | 50 (40–60) |
| Electricity demand | $kWh_{el}/kg_{H2}$ | both | 1.7 (1.5–1.9) | 2.8 (2.5–3.2) | 0.6 | 1.7 (1.5–1.9) |
| Filling type | – | both | 700 bar | 700 bar | 700 bar | liquid |
| $H_2$ loss[b] | % | both | 0.5 | 0.5 | 1.2 (0.5–1.9) | 1.7 (1.0–2.4) |

[a] Refers to the maximum hourly capacity.
[b] Percentage of hydrogen losses per unit hydrogen filled and depends (especially for the liquid systems) strongly on the capacity rate.

evaporation and cooler are made. Each filling station is designed to provide the maximum hourly filling capacity for 5 h. Afterward, the power is reduced to the compressor, respectively evaporator limitations to 65% of the maximum filling capacity. The maximum filling capacity of 320 $kg_{H2}$/h is based on the assumption that four trucks with a tank capacity of 80 kg of hydrogen each could be filled within 1 h. Based on component-specific techno-economic data [85] the overall filling station design specific techno-economic parameters are calculated and listed in Table 11.

## Results and discussion

Below, the results of the assessment of the introduced hydrogen supply chains are presented and discussed for the years 2030 and 2050. Due to the fact that the overall hydrogen supply cost are divided into several segments in the upcoming figures, these segments declarations are summarized as follows.

- H2 production – excl losses. Defines the specific hydrogen production cost in the respective region (levelized cost of hydrogen) and does not vary for the same production regions.
- H2 production – only losses. Defines the additional specific hydrogen production cost due to an additional hydrogen demand caused by supply chain losses (e.g., boil-off losses, purification losses).
- Storage. Defines the sum of specific costs of all storage processes (e.g., buffer storage, harbor storage, seasonal storage) needed within the specific hydrogen supply chain.
- Conversion – excl carrier. Defines the sum of the specific costs of all conversion processes (e.g., compression, ammonia synthesis) being part within the specific hydrogen supply chain; only the specific cost for the hydrogen carrier (e.g., dibenzyltoluene for liquid organic hydrogen carriers, carbon dioxide for methanol) is considered separately. In the case of the liquid hydrogen filling, liquefaction is not included here (see segment Liquefaction).
- Conversion – only carrier. Defines the specific cost caused by the needed hydrogen carrier and thus only appears in the liquid organic hydrogen carrier, methanol and ammonia hydrogen supply chains.
- Liquefaction (central). Only occurs in the case of the liquid hydrogen filling and defines the specific cost for centralized liquefaction.
- LH2 pumps. Defines the sum of the specific costs of all liquid hydrogen pumps occurring in the respective supply chain.

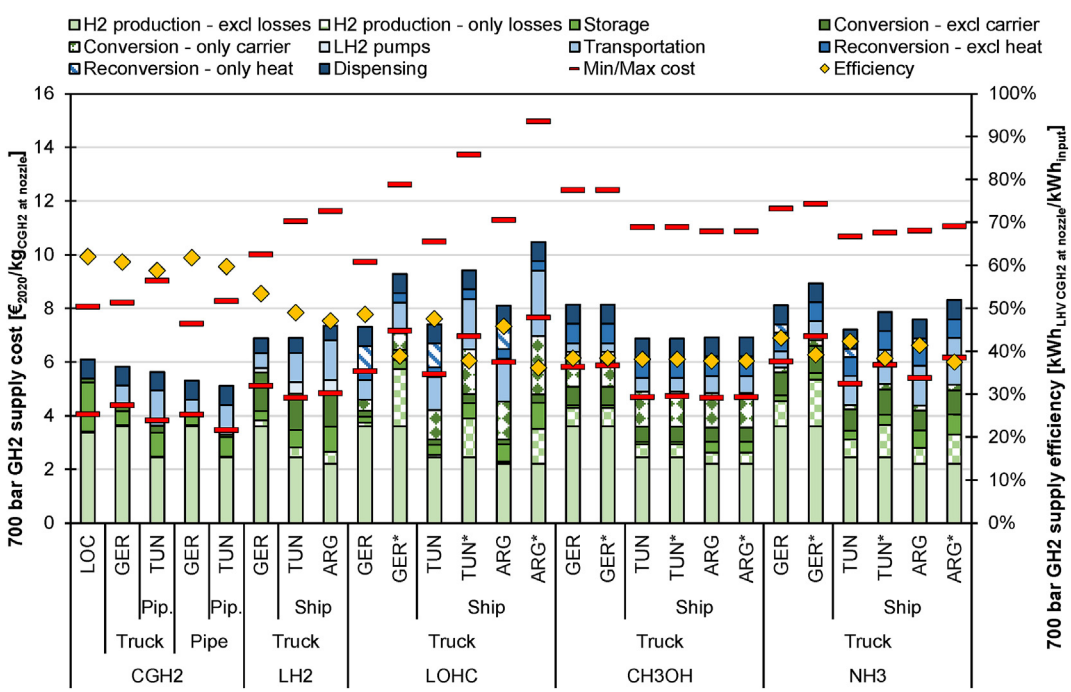


**Fig. 4 – Supply cost and efficiency from well to tank in 2030 for compressed gaseous hydrogen filling (700 bar GH2) of heavy-duty vehicles in Germany (* = heat demand for the reconversion is supplied internally by using the needed energy fraction from the released hydrogen; ARG = hydrogen production in Argentina (Patagonia); CGH2 = compressed gaseous hydrogen supply chain; CH3OH = methanol supply chain; GER = centralized production in the North Sea of Germany; LH2 = liquid hydrogen supply chain; LOC = local production directly at the filling station in Central Germany; LOHC = liquid organic hydrogen carrier supply chain; NH3 = ammonia supply chain; Pip. = pipeline import; Pipe = pipeline distribution to filling station; Ship = ship import; TUN = hydrogen production in Tunisia; Truck = truck distribution to filling station).**

- Transportation. Defines the sum of the specific cost of all transportation processes (e.g., ship import and truck distribution) of the considered supply chain.
- Reconversion – excl heat. Defines the specific cost of the reconversion process (e.g., ammonia cracking) without the specific cost for the needed heat (own segment).
- Reconversion – only heat. Defines the specific cost caused by the externally supplied thermal energy needed for the reconversion; in case of an assumed internal hydrogen use, the additional cost caused by the reconversion heat demand is considered in the additional hydrogen production (H2 production – only losses).
- Liquefaction (decentral). Only occurs in the case of the liquid hydrogen filling and defines the specific cost for decentralized liquefaction directly at the filling station.
- Dispensing. Defines the specific cost for the compressed gaseous respectively liquid hydrogen filling.
- Min/Max cost. Defines the overall hydrogen supply cost for the progressive (minimum) and conservative (maximum) scenarios.
- Efficiency. Defines the overall efficiency of the specific hydrogen supply chain for the baseline scenario.

### *Compressed gaseous hydrogen filling*

In the case of compressed gaseous hydrogen filling, hydrogen is needed at the end of the supply chain at a pressure level of 700 bar to be dispensed into the vehicle. The respective assessment results of the different hydrogen supply chains are shown in Fig. 4 for the year 2030.

Thus, the hydrogen production costs dominate the overall hydrogen supply cost. They are between almost 2 €$_{2020}$/kg$_{H2}$ for hydrogen production in Argentina (ARG) and up to 4 €$_{2020}$/kg$_{H2}$ for hydrogen produced in North Germany (GER) based on offshore wind power. Additionally, the costs for conditioning, transportation and storage vary significantly for the different supply chains.

The lowest overall supply costs result for the supply chains where hydrogen is transported and distributed as compressed gaseous hydrogen (CGH2) in pipelines directly to the filling station leading to hydrogen supply cost down to 5 €$_{2020}$/kg$_{H2}$ for the baseline scenario with a bandwidth between 4 and 8 €$_{2020}$/kg$_{H2}$ (progressive and conservative scenario). Furthermore, the overall costs of an import from Tunisia (TUN) per pipeline are slightly lower compared to a centralized hydrogen production in North Germany.

In the case of a local hydrogen production directly at the filling station (LOC), the hydrogen is stored in a pressure tank and not in a salt cavern. Therefore, although no transportation is needed, the hydrogen supply cost (roughly 6 €$_{2020}$/kg$_{H2}$) is slightly higher than for the other compressed gaseous hydrogen supply chains. The efficiencies of the different hydrogen supply chains are comparatively high (around 60%).

The second and third lowest cost options are hydrogen supply chains using liquid hydrogen (LH2) or methanol (CH3OH) as a hydrogen carrier, showing a cost increase of roughly 40% and thus leading to hydrogen supply cost of around 7 €$_{2020}$/kg$_{H2}$. Also, the bandwidth is higher and varies between 5 and 12 €$_{2020}$/kg$_{H2}$. Therefore, liquefaction leads to a significant cost increase compared to the gaseous supply chains. In the case of an import, additional cost drivers are boil-off losses, cryo-pumps and sea transportation – and here especially for long import distances. Hence, an import from Tunisia to Germany leads to lower hydrogen supply costs than an import from Argentina (Patagonia). Compared to the gaseous supply chains the overall efficiencies for the liquid hydrogen supply chains are lower due to the high energy demand for liquefaction, leading to overall efficiencies of ca. 50%.

Considering methanol as a derivate, the cost for the carbon dioxide supply – direct air capture is assumed for the baseline scenario – and the cost for the reconversion are significant additional cost drivers. Thereupon, hydrogen transportation costs by using methanol as a hydrogen carrier are low and therefore suitable for long-distance transport. This leads to the fact that an import from Tunisia and Argentina (Patagonia) shows lower hydrogen supply cost than the production in Germany itself. Since the high purification losses of hydrogen are assumed to be used as an energy source for the heat supply, the net heat energy demand is also zero for the external heat supply option and no variation in cost or efficiency occurs. Regarding efficiency, the high energy demand for the carbon dioxide capture and the reconversion of methanol results in a lower overall efficiency (40%) compared to the liquid hydrogen supply chains.

For liquid organic hydrogen carriers and ammonia, the hydrogen supply costs are in a magnitude of 7–8 €$_{2020}$/kg$_{H2}$ for the baseline scenario varying between 6 and 11 €$_{2020}$/kg$_{H2}$ if the reconversion heat is supplied by an external source.

In the case of liquid organic hydrogen carrier, a main cost driver is the heat needed for reconversion. Furthermore, the cost for the carrier, dibenzyltoluene, is especially important for long storage and transportation times leading to a high amount of carrier bound within the supply chain and therefore tied capital. Again, the overall hydrogen supply efficiency is lowered due to the heat energy. In comparison to methanol, the energy demand for dibenzyltoluene production is neglectable since the carrier is recirculated. Nevertheless, heat for the reconversion is still needed resulting to an overall chain energy efficiency of less than 50%.

By using ammonia as a derivate, the cost for the nitrogen as a hydrogen carrier is of minor importance compared to the costs for conversion and reconversion. However, the most crucial part is the hydrogen purification loss after ammonia is cracked again into nitrogen and hydrogen. This results in a high additional hydrogen production demand and, in the end, to a significant increase of the hydrogen supply cost. Compared to the liquid organic hydrogen carrier, the cost for the reconversion heat are of minor importance because the specific energy demand for the hydrogen release is lower and additionally, the hydrogen purification losses are assumed to be used for a heat generation. Due to the hydrogen losses discussed above, the overall efficiency is lower compared to the liquid organic hydrogen carrier chains and overall under 45%.

If the necessary reconversion heat is supplied internally by using the released hydrogen, a significant increase of the additional produced hydrogen demand occurs. Furthermore, also this additional hydrogen demand needs to be

conditioned, stored and transported to fulfill the pre-defined demand at the filling station. These circumstances lead to a strong increase of the hydrogen supply cost and a decrease of efficiency, especially for liquid organic hydrogen carriers since they obtain the highest specific heat demand for the reconversion.

In Fig. 5, the hydrogen supply cost and efficiency for the year 2050 are visualized. The general relation between the different hydrogen supply chains does not change tremendously. Still, hydrogen production cost has the highest share related to the overall cost, even if significant cost reductions for photovoltaic, wind power and electrolyzers are taken into account. The compressed gaseous hydrogen supply chain still appears clearly as the most promising option under the assumption taken here. For the baseline scenario, the costs are around 4 $€_{2020}/kg_{H2}$ and efficiencies up to 65%. In the case of liquid hydrogen and methanol, hydrogen supply costs are close to 5 $€_{2020}/kg_{H2}$, while in the case of liquid organic hydrogen carriers and ammonia costs are around 6 $€_{2020}/kg_{H2}$.

Additional information regarding the cost distribution for the progressive and conservative scenarios, as well as for the energy input distribution for the baseline scenarios, are presented in the supplementary material A. 2 to A. 7.

### *Liquid hydrogen filling*

When considering liquid hydrogen filling, liquefied hydrogen at −253 °C is needed before it will be filled into the vehicle. Hence, hydrogen needs to be liquefied before the filling process into the HDV if it is not supplied as liquid hydrogen anyway. The results of the energetic and economic assessment of the different hydrogen supply chains under these given circumstances are shown in Fig. 6 for the year 2030.

The specific cost shares of the different chain sections are still similar to the previously described relations of the gaseous hydrogen filling supply chains. For example, hydrogen production is still a main cost driver for each supply scenario. However, also liquefaction, especially if a small scale liquefaction is assumed, is contributing significantly to the overall supply cost.

Thus not a gaseous hydrogen supply chain per pipeline connection, but the liquid hydrogen supply chains obtain the lowest hydrogen supply costs with approximately 7 $€_{2020}/kg_{H2}$ with a bandwidth between 5 and 12 $€_{2020}/kg_{H2}$ and efficiencies around 50% depending on the transportation distance. In addition, also a local hydrogen production, liquefaction and storage directly at the filling station shows low supply cost of around 7 $€_{2020}/kg_{H2}$ (bandwidth between 5 and 12 $€_{2020}/kg_{H2}$) and a relative high overall energetic efficiency (55%), since the cost for the seasonal storage of liquid hydrogen is significantly lower compared to a gaseous hydrogen storage within a pressure tank. The supply costs of the other gaseous hydrogen supply chains are slightly higher (8 $€_{2020}/kg_{H2}$), depending on the delivery type per pipeline or truck.

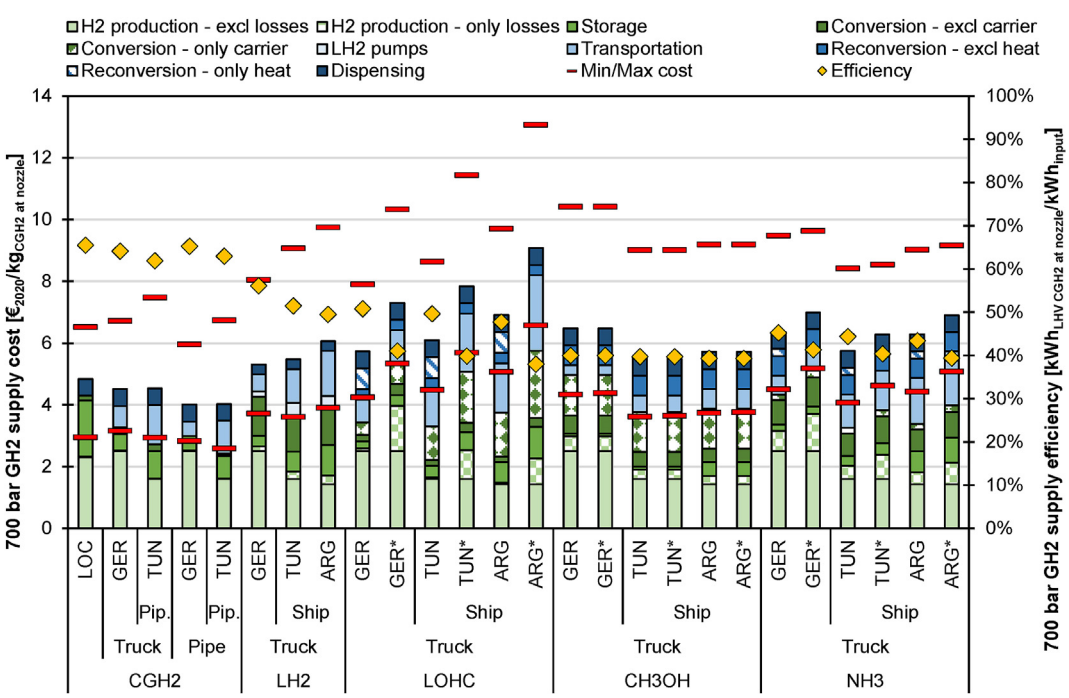


**Fig. 5 – Supply cost and efficiency from well to tank in 2050 for compressed gaseous hydrogen filling (700 bar GH2) of heavy-duty vehicles in Germany (* = heat demand for the reconversion is supplied internally by using the needed energy fraction from the released hydrogen; ARG = hydrogen production in Argentina (Patagonia); CGH2 = compressed gaseous hydrogen supply chain; CH3OH = methanol supply chain; GER = centralized production in the North Sea of Germany; LH2 = liquid hydrogen supply chain; LOC = local production directly at the filling station in Central Germany; LOHC = liquid organic hydrogen carrier supply chain; NH3 = ammonia supply chain; Pip. = pipeline import; Pipe = pipeline distribution to filling station; Ship = ship import; TUN = hydrogen production in Tunisia; Truck = truck distribution to filling station).**

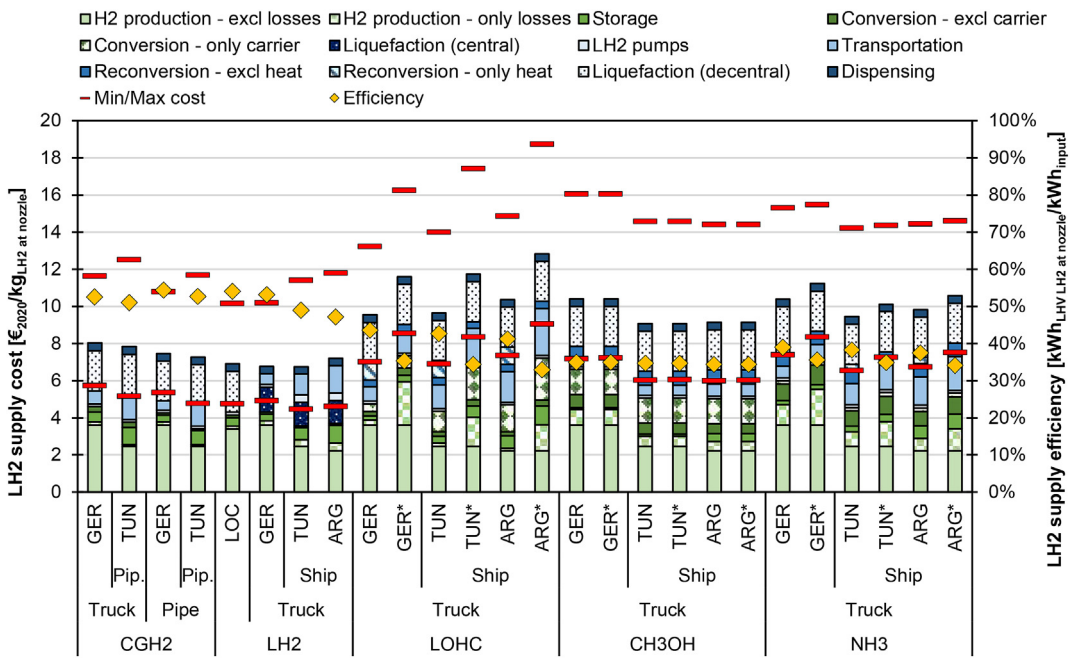


**Fig. 6 – Supply cost and efficiency from well to tank in 2030 for liquid hydrogen filling (LH2) of heavy-duty vehicles in Germany (* = heat demand for the reconversion is supplied internally by using the needed energy fraction from the released hydrogen; ARG = hydrogen production in Argentina (Patagonia); CGH2 = compressed gaseous hydrogen supply chain; CH3OH = methanol supply chain; GER = centralized production in the North Sea of Germany; LH2 = liquid hydrogen supply chain; LOC = local production directly at the filling station in Central Germany; LOHC = liquid organic hydrogen carrier supply chain; NH3 = ammonia supply chain; Pip. = pipeline import; Pipe = pipeline distribution to filling station; Ship = ship import; TUN = hydrogen production in Tunisia; Truck = truck distribution to filling station).**

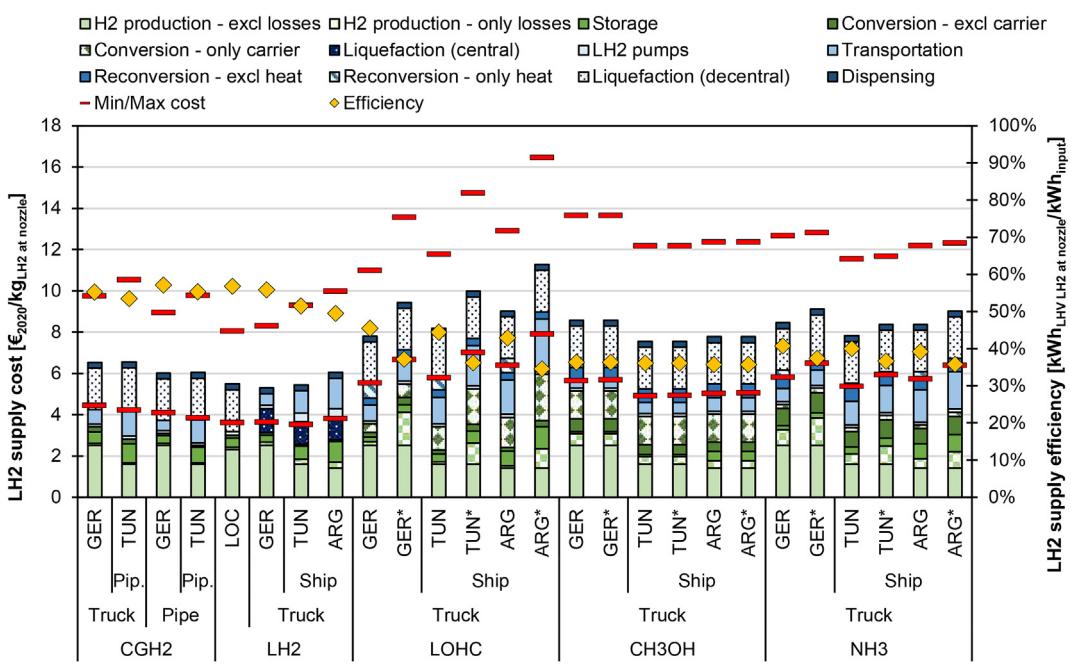


**Fig. 7 – Supply cost and efficiency from well to tank in 2050 for liquid hydrogen filling (LH2) of heavy-duty vehicles in Germany (* = heat demand for the reconversion is supplied internally by using the needed energy fraction from the released hydrogen; ARG = hydrogen production in Argentina (Patagonia); CGH2 = compressed gaseous hydrogen supply chain; CH3OH = methanol supply chain; GER = centralized production in the North Sea of Germany; LH2 = liquid hydrogen supply chain; LOC = local production directly at the filling station in Central Germany; LOHC = liquid organic hydrogen carrier supply chain; NH3 = ammonia supply chain; Pip. = pipeline import; Pipe = pipeline distribution to filling station; Ship = ship import; TUN = hydrogen production in Tunisia; Truck = truck distribution to filling station).**

Accordingly, the hydrogen supply chains using another carrier molecule, namely the liquid organic hydrogen carriers, methanol and ammonia, show partially significant higher supply costs of 9 $€_{2020}/kg_{H2}$ and more, even if the needed reconversion heat is not supplied internally by a fraction of the released hydrogen. Furthermore, the efficiencies are clearly lower compared to the other supply chains.

In the case of 2050 as a reference year (Fig. 7), the liquid hydrogen supply cost is calculated to decrease to under 6 $€_{2020}/kg_{H2}$ for the baseline scenario for the liquid hydrogen supply chain, obtaining an overall efficiency of roughly 55%. However, the general relations between the different hydrogen options have not changed significantly and the compressed gaseous and liquid hydrogen supply chains still show lower hydrogen supply costs and efficiencies than chains using liquid organic hydrogen carriers, ammonia or methanol as hydrogen carriers. Additional information can be found in the supplementary material in A. 8 to A.13.

## Conclusion

The overall goal of this paper is to evaluate and compare the energy efficiency and cost of a green hydrogen supply from well to tank for hydrogen fuelled HDV in Germany for the years 2030 and 2050. To do so, different hydrogen supply chains are defined by assuming Central Germany, North Germany, Tunisia and Argentina (Patagonia) as hydrogen production locations and considering compressed gaseous hydrogen (CGH2), liquid hydrogen (LH2), liquid organic hydrogen carriers (LOHC), methanol (CH3OH) and ammonia (NH3) as conditioned hydrogen options for transportation and storage.

The presented results identify that transportation and storage of hydrogen in its state of aggregation as it is demanded results in the lowest supply costs at the dispenser; a compressed gaseous hydrogen supply chain shows the lowest supply cost for compressed gaseous hydrogen filling and liquid hydrogen supply chains for liquid hydrogen filling. Overall, the following conclusions can be drawn for the different hydrogen supply options assessed here.

- **Compressed gaseous hydrogen supply chains** show low hydrogen supply costs if a direct pipeline connection to the filling station can be realized, while a hydrogen transport per truck leads to a cost increase. In the case of compressed gaseous hydrogen filling, they result into lowest overall hydrogen supply costs when producing hydrogen centralized in North Germany or importing it from Tunisia. The lower hydrogen production cost in Tunisia due to a lower cost renewable energy supply compared to Germany are neutralized by the transportation costs in comparison to hydrogen production based on offshore wind power in North Germany. Additionally, a low-cost compressed gaseous hydrogen storage (e.g., in a salt cavern) is crucial for the overall supply costs since an alternatively to that realized seasonal storage in pressure tanks leads to significantly higher costs, as is the case of the local, decentralized hydrogen production scenario directly at the filling station.
- **Liquid hydrogen supply chains** are characterized by their high capital expenditures and energy demand for the liquefaction as well as its boil-off losses, leading to higher gaseous hydrogen supply costs compared to the gaseous supply chains. However, in the case of a liquid hydrogen demand by the final consumer (here: HDV), a liquid hydrogen supply chain is the most favorable option. Liquid hydrogen transportation over long distances, e.g., from Argentina (Patagonia), is non-beneficial due to the resulting significant boil-off losses. This is even true if the hydrogen production costs are the lowest. Therefore, liquid hydrogen is a more promising option for short import distances (e.g., from Tunisia). Again, the overall supply cost between a centralized hydrogen production based on offshore wind power in North Germany and an import from Tunisia are in the same order of magnitude.
- **Liquid organic hydrogen carrier supply chains** show the highest hydrogen supply costs due to the high energy demand for dehydrogenation and the relatively low reversible hydrogen capacity of 5.6 wt%. In addition, the capital expenditures for the carrier molecule increases clearly with the transportation distance and the storage time. Hence, even if liquid organic hydrogen carriers are easy to handle within the existing infrastructure, it is not preferable to transport them over long distances or use them as a seasonal storage option. If the released hydrogen is also used for the heat supply needed for reconversion, overall supply costs increase even further. Is waste heat (>300 °C) available at the reconversion site, or the generated heat at the conversion location can achieve a monetary value, the supply cost can be lowered. Nevertheless, even then this chain shows still not the lowest supply costs. This is also true for methanol and ammonia.
- **Methanol supply chains** are mainly determined by the costs of the carbon dioxide and the energy demand for the reconversion. Thereupon, the specific hydrogen transportation and storage cost are low. Therefore, it is suitable for long-distance imports (e.g., from the South of South America). Methanol appears as an overall lower cost conditioned hydrogen option compared to liquid organic hydrogen carriers and ammonia, but still worse than compressed gaseous and liquid hydrogen supply chains if pure hydrogen should be supplied at the filling station. However, local biogenic point sources could lead to lower carbon dioxide supply cost and consequently, the overall hydrogen supply cost can be reduced too.
- **Ammonia supply chains** are characterized by relatively high costs for the conditioning processes and especially the very high hydrogen purification losses, leading to high additional hydrogen production, transportation and storage demand, are the main cost drivers. Thereupon, the cost of nitrogen, used as the hydrogen carrier, is compared to liquid organic hydrogen carrier and methanol, from minor importance.

By comparing the different hydrogen production regions, even if Patagonia (Argentina) shows the lowest hydrogen production cost, the hydrogen supply costs of the different pathways are higher due to the long importing distance. Compared to that, the hydrogen supply costs of an import from Tunisia are in the same order of magnitude as a centralized production in North Germany based on offshore wind power. If onshore wind power is considered for North Germany, supply cost for

this region will drop clearly. However, due to the given space constraints, large-scale hydrogen production by using onshore wind power seems to be questionable. The local hydrogen production directly at the filling station is, in general, not advantageous for compressed gaseous filling due to the high cost for the necessary pressure tank to harmonize weather dependent production and customer related demand. In the case of liquid filling, local hydrogen production and liquefaction can be economically attractive and reach similar costs as the liquid hydrogen supply chains.

For the most favorable supply chain options compressed gaseous filling costs are determined to be around 5 €$_{2020}$/kg$_{H2}$ (bandwidth around 4 to 8 €$_{2020}$/kg$_{H2}$) in 2030 and overall energetic efficiency sums up to roughly 60%. For the year 2050, further cost reductions are expected, leading to supply costs of 4 €$_{2020}$/kg$_{H2}$ (bandwidth of 3–7 €$_{2020}$/kg$_{H2}$) and energetic efficiencies up to 65%. On the contrary, the costs for liquid hydrogen filling are estimated higher with 7 €$_{2020}$/kg$_{H2}$ (bandwidth between 5 and 12 €$_{2020}$/kg$_{H2}$) and the overall efficiency is calculated with ca. 50% in 2030. For 2050 supply costs expected to drop to under 6 €$_{2020}$/kg$_{H2}$ (bandwidth between 4 and 8 €$_{2020}$/kg$_{H2}$) and the efficiency increases to 55%.

## CRediT authorship contribution statement

L. Sens: Conceptualization, Data curation, Investigation, Methodology, Validation, Visualization, Writing – original draft. U. Neuling: Methodology, Writing – review &editing, Supervision. K. Willbrand: Conceptualization, Validation, Writing – review &editing. M. Kaltschmitt: Conceptualization, Writing – review &editing, Supervision.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijhydene.2022.07.113.

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